

## 1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

**2. LEVEL 0 DFD (CONTEXT DIAGRAM)** This diagram illustrates the macro-level interactions of the ComicCraft system.

- **Input Entity (End-User):** Submits [Story Prompts & Stylistic Choices].
- **Core Application (ComicCraft System):** Processes the inputs, coordinates with external models, and delivers a [Browser Preview] and a [Downloadable PDF].
- **External Entity 1 (Google Gemini LLM):** Receives the [Contextual Prompts] and returns a [Structured JSON Outline & Script].
- **External Entity 2 (Hugging Face API):** Receives the [Scene Descriptions] and returns [Image Byte Data].

**3. LEVEL 1 DFD (SYSTEM PROCESS FLOW)** This section breaks down the sequential internal logic of the backend application.

- **Process 1.0 (Data Capture):**
  - Captures HTTP form submissions.
  - *Flow:* HTML Form → FastAPI Request Object.
- **Process 2.0 (Narrative Engineering):**
  - Transmits data to Gemini 1.5 Flash and Pro to build the story.
  - *Flow:* Validated Prompt → 5-Panel JSON Sequence & Dialogue Array.
- **Process 3.0 (Image Synthesis):**
  - Iterates through the JSON sequence, pinging Stable Diffusion for graphics.
  - *Flow:* Scene Prompts → .jpg files saved in the local static directory.
- **Process 4.0 (Content Assembly):**
  - Maps the synthesized text to the corresponding visual files.
  - *Flow:* Texts + Image Paths → Unified Layout Dictionary.
- **Process 5.0 (PDF Export):**

- Feeds the layout dictionary into the FPDF layout engine.
- *Flow*: Layout Dictionary → Compiled comic\_export.pdf.